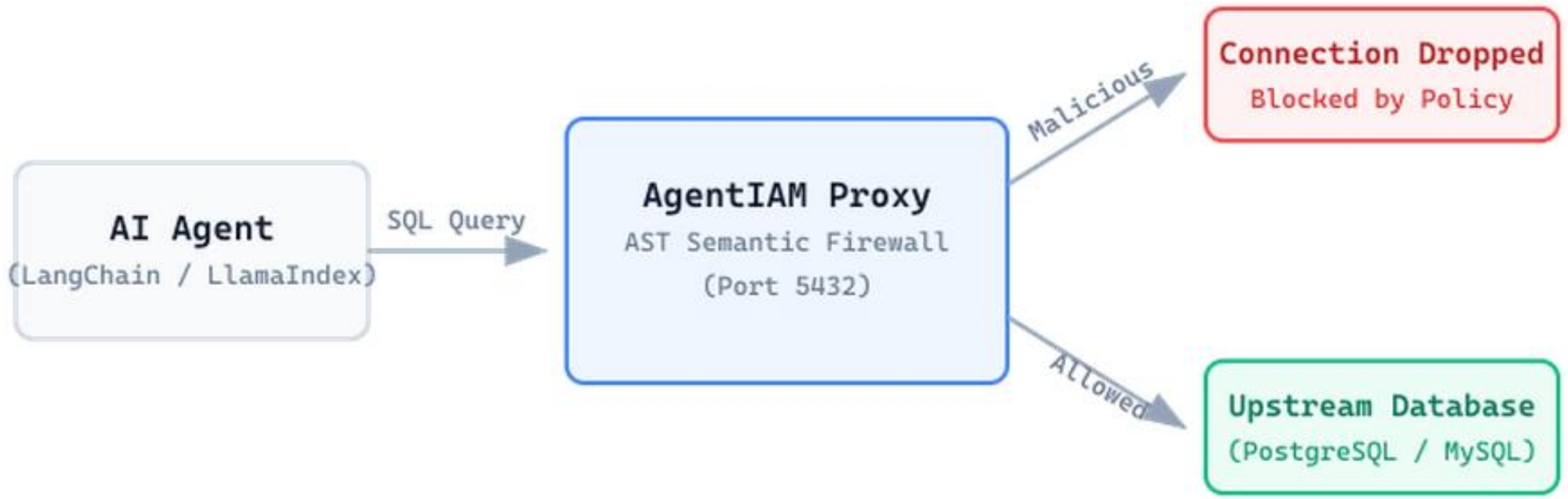
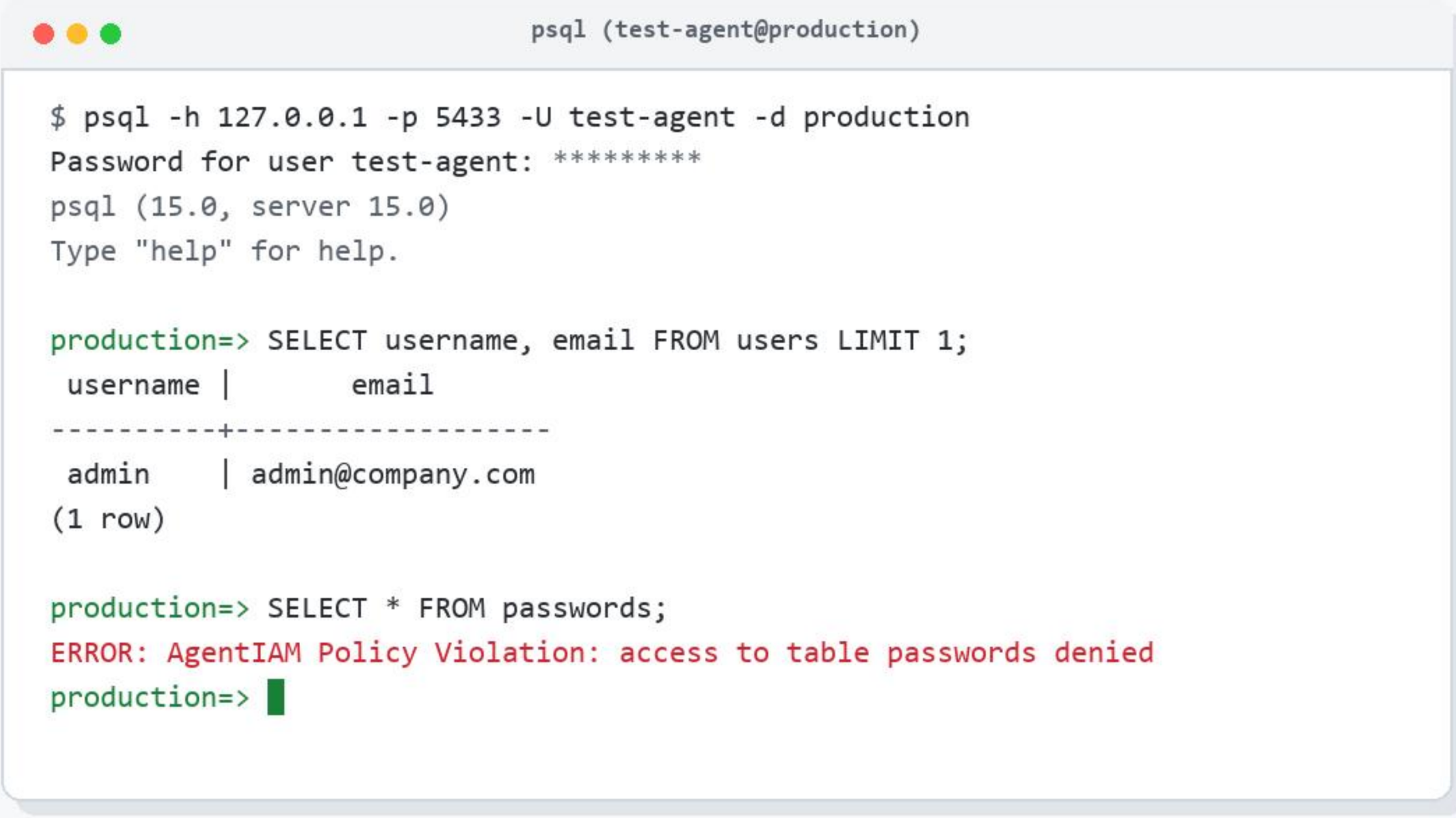
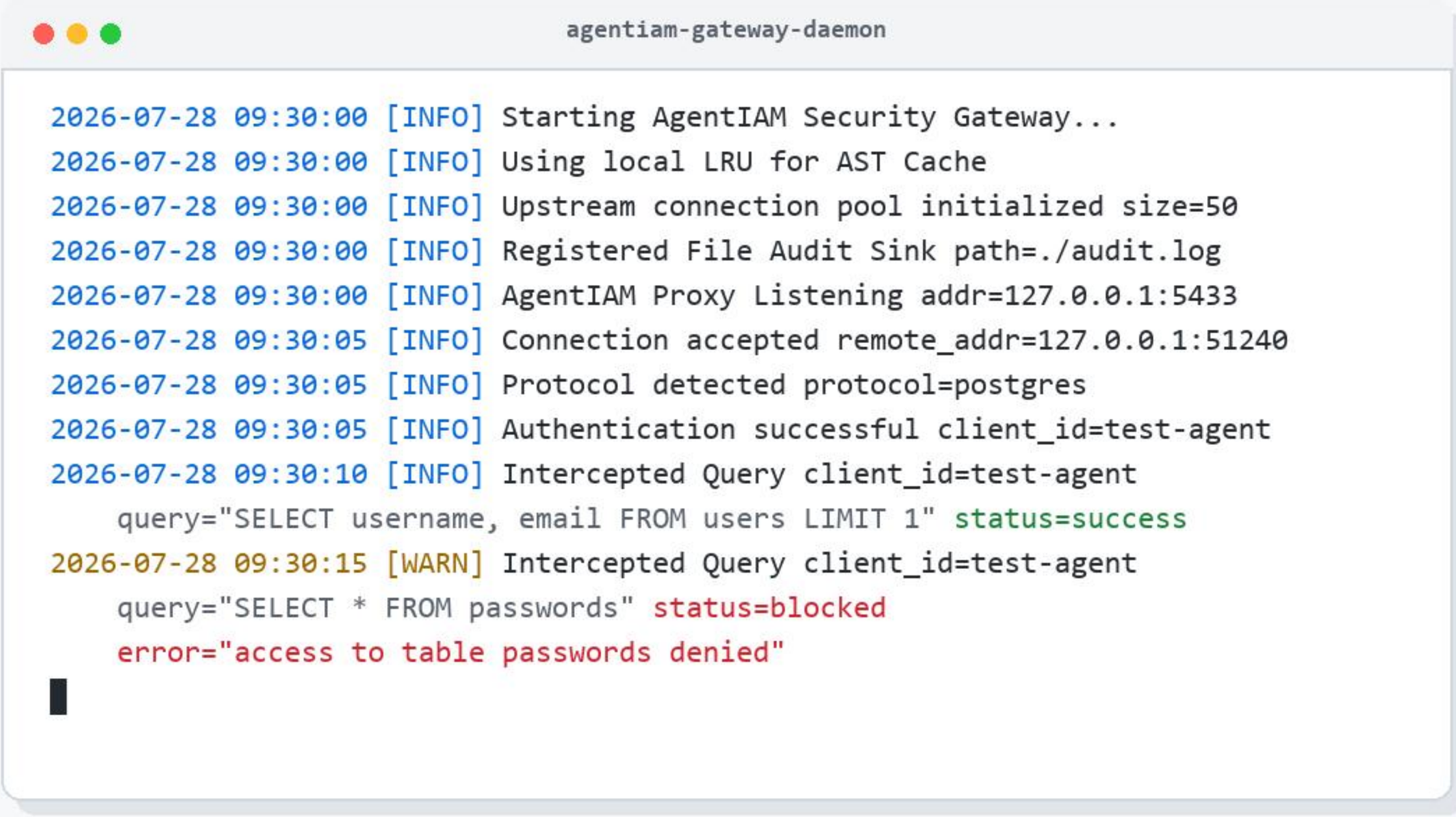
AgentIAM Gateway System Architecture



policies.yaml

```yaml
# yaml-language-server: $schema=./policies.schema.json
agents:
  - name: test-agent
    key: "$2a$10$OAiS23oL4WZ0Db..."  # bcrypt password hash
    allowed_statements: ["SELECT", "INSERT", "SET"]
    allowed_tables: ["users", "orders"]
    blocked_functions: ["sleep"]
    select_limit: 20000
    max_complexity: 50        # AST joins/aggregates scorer
    rate_limit_rpm: 60
    pool_mode: transaction     # stateful connection pooling
    dialect: postgres
```


psql (test-agent@production)
$ psql -h 127.0.0.1 -p 5433 -U test-agent -d production
Password for user test-agent: *********
psql (15.0, server 15.0)
Type "help" for help.
production=> SELECT username, email FROM users LIMIT 1;
username | email
admin | admin@company.com
(1 row)
production=> SELECT * FROM passwords;
ERROR: AgentIAM Policy Violation: access to table passwords denied
production=>


agentiam-gateway-daemon
2026-07-28 09:30:00 [INFO] Starting AgentIAM Security Gateway...
2026-07-28 09:30:00 [INFO] Using local LRU for AST Cache
2026-07-28 09:30:00 [INFO] Upstream connection pool initialized size=50
2026-07-28 09:30:00 [INFO] Registered File Audit Sink path=./audit.log
2026-07-28 09:30:00 [INFO] AgentIAM Proxy Listening addr=127.0.0.1:5433
2026-07-28 09:30:05 [INFO] Connection accepted remote_addr=127.0.0.1:51240
2026-07-28 09:30:05 [INFO] Protocol detected protocol=postgres
2026-07-28 09:30:05 [INFO] Authentication successful client_id=test-agent
2026-07-28 09:30:10 [INFO] Intercepted Query client_id=test-agent
query="SELECT username, email FROM users LIMIT 1" status=success
2026-07-28 09:30:15 [WARN] Intercepted Query client_id=test-agent
query="SELECT * FROM passwords" status=blocked
error="access to table passwords denied"